

Accuracy in Biometric Systems

- The key performance metrics in biometrics are
 - False match rate, (FMR)
 - False non- match rate (FNMR)
 - Failure-to-enroll rate (FER)

Quick Refs.

1. False Match Rate (FMR) / FAR

Measures the proportion of **impostor attempts** that are **incorrectly accepted**.

$$FMR = FAR = \frac{\text{Number of false matches}}{\text{Total impostor comparison attempts}} \times 100$$

2. False Non-Match Rate (FNMR)

Measures the proportion of **genuine attempts** that are **incorrectly rejected**.

$$FNMR = \frac{\text{Number of false non-matches}}{\text{Total genuine comparison attempts}} \times 100$$

3. Failure-to-Enroll Rate (FER)

Measures the proportion of users who **cannot be successfully enrolled**.

$$FER = \frac{\text{Number of failed enrollments}}{\text{Total enrollment attempts}} \times 100$$

False match rate, (FMR)

- **Definition :** A biometric solution's false match rate is the probability that a user's template will be incorrectly judged to be a match for a different user's template.
- False matches **may occur because two people have similar enough biometric characteristics**
- The false match rate is often referred to as the **false acceptance rate (FAR)**.



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False match rate, (FMR)

- False match rates **can be reduced by adjusting thresholds** that adjust the level of correlation necessary for two templates to be judged a match.
- **False match rate and false nonmatch rate are inversely related** and must always be assessed.
- Ex: A system with a false match rate of 0 percent, but a false nonmatch rate of 50 percent, **is secure but unusable**.



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False match rate, (FMR)

- The false match rate is often referred to as the **false acceptance rate (FAR)**.
- The term false acceptance rate assumes that the result of a successful match is that the user is accepted into a building, an application, or a resource, which is often the case.
- The false match rate is often referred to as the **false acceptance rate (FAR)**.

When Are False Matches Acceptable?

- A false match does not always mean that the biometric system has failed or needs threshold adjustment.
- **Real Scenario** :Think of a video security system used by a casino to identify known card counters.

Solution :

- **Without Biometrics** : security personnel need to memorize the faces of every known card counter, then look at every face at every blackjack table.

When Are False Matches Acceptable?

Solution :

- **With Biometrics** : facial-scan software can scan the images provided by security cameras, locate faces, and rapidly perform 1:N identification against databases of card counters.
- The facial-scan system will be programmed to notify or flag any faces that come at all close to any images in the card counter database, at which point the security personnel can determine whether the two faces—one in the database, one from the live security camera—are a match. **Even if 9 out of 10 matches are incorrect**, the facial-scan system is **performing a valuable function by alerting staff to possible matches and reducing overall manual effort**.

False Match Rate in Large-Scale Identification Systems

- FMR are often different for large-scale identification systems. Because over a billion matches might take place on a given day
- large-scale identification requires much lower false match rates than would commonly be the case in verification systems.
- In particular, negative identification systems—those designed to ensure that an individual's biometric data is not present more than once in a database require low false match rate

False Nonmatch Rate(FNMR)

- (FNMR) is the probability that a user's template will be incorrectly judged to not match his or her enrollment template.
- In most cases, a false nonmatch means that an authorized user is locked out of a system, incorrectly denied access to a facility or resource.
- A solution's false nonmatch rate is often referred to as false rejection rate (FRR).

False Nonmatch Rate(FNMR)

- False nonmatches occur because there is **not a sufficiently strong correlation** between a user's verification and enrollment templates.

Main Reason for FNMR:

- Changes in a user's biometric data
- Changes in how a user presents biometric data
- Changes in the environment in which data is presented

Changes in a User's Biometric Data

- Many biometric technologies are based on **characteristics that can change between enrollment and verification, especially as a substantial amount of time elapses.**
- Examples :
 - **Voice-scan** systems, for example, do not perform as well when users have sore throats;
 - **facial-scan** systems perform poorly if a user has experienced drastic changes in facial hair or weight.

Changes in a User's Biometric Data

- Even more stable characteristics such as **fingerprints can change over time, as scars, aging, and general wear and tear** lower the quality of the data.
- Technologies somewhat susceptible to changes in underlying biometric data include facial-scan and hand-scan; eye-based biometrics, iris-scan, and retinal scan are less susceptible. Long-term robustness of finger-scan solutions varies too widely from vendor to vendor to make a technology-wide assessment.

Changes in User Presentation

- Changes in User Presentation Even if the biometric data has not changed in any way, the manner in which a **user presents data to a biometric system can lead to substantial false rejection rates.**
- These changes are not necessarily attributable to user carelessness.
- **Finger-scan systems** may advise users to move a finger up or down;
- **voice-scan systems** may advise users to speak more softly or loudly.
- Behavioral biometrics, in particular, are subject to changes in user presentation.

Changes in Environment

- changes in authentication environment.
 - Variations in **background lighting and composition, noise, and even temperature can lead to increased false non- match rates.**

Example :

Enrolling on a landline telephone, for example, and attempting to verify via mobile phone is likely to lead to high FNMR.

Factors Affecting False Nonmatch Rates

TECHNOLOGY	SUSCEPTIBILITY TO CHANGES IN BIOMETRIC DATA	SUSCEPTIBILITY TO CHANGES IN USER PRESENTATION	SUSCEPTIBILITY TO ENVIRONMENTAL CHANGES
Finger-scan	<i>Low to moderate.</i> Some user groups are prone to changes in fingerprint quality; for most users, the fingerprint remains stable and usable.	<i>Moderate.</i> Variation in angle, pressure, and position can lead to high FNMR.	<i>Low.</i> Some solutions are susceptible to light and temperature changes, but this is rare.
Facial-scan	<i>Moderate to high.</i> Changes in facial hair and hairstyle, as well as drastic changes in weight, can increase FNMR.	<i>Low to moderate.</i> In a controlled environment, user presentation should not vary. Otherwise, user is responsible for quality of presentation. Removal of glasses and hats can also increase FNMR.	<i>Moderate to high.</i> Changes in intensity and angle of lighting, as well as background composition, can be highly problematic for facial-scan.
Voice-scan	<i>Low to moderate.</i> Illnesses can affect the voice to the point where FNMR increases.	<i>Moderate to high.</i> Volume of speech, inflection, and duration must be consistent to ensure verification.	<i>Moderate to high.</i> Background noise and quality of acquisition device (microphone, telephone) can increase FNMR.

Factors Affecting False Nonmatch Rates

Iris-scan	<i>Low.</i> The iris is highly stable and, short of eye trauma, iris patterns do not change.	<i>Moderate to high.</i> Users must be positioned correctly in order for high-quality image to be captured.	<i>Low.</i> Lighting does not affect the technology, as it uses infrared illumination.
Signature-scan	<i>Moderate to high.</i> Data and presentation cannot be effectively separated. For some users, the signature is highly susceptible to variation, especially over time. Signing in different position (sitting vs. standing) may impact performance.		<i>Low.</i> Technology largely unaffected by environment.
Hand-scan	<i>Moderate.</i> Injuries, swelling, water retention can increase likelihood of false nonmatching.	<i>Low to moderate.</i> Design of technology facilitates correct placement, but placement itself can be challenging for some users.	<i>Low.</i> Technology largely unaffected by environment.

TECHNOLOGY	SUSCEPTIBILITY TO CHANGES IN BIOMETRIC DATA	SUSCEPTIBILITY TO CHANGES IN USER PRESENTATION	SUSCEPTIBILITY TO ENVIRONMENTAL CHANGES
Retina-scan	<i>Low.</i> Aside from dislocation due to age-related eye diseases, retinal patterns are stable.	<i>High.</i> User presentation is extremely challenging, leading to atypically high FNMR.	<i>Low.</i> Technology largely unaffected by environment.
Keystroke-scan	<i>Moderate to high.</i> Data and presentation cannot be effectively separated. Typing patterns easily change, especially for poor typists. Effect of different keyboard types (click vs. nonclick, ergonomic keyboards) unknown.		<i>Low.</i> Technology largely unaffected by environment.

Failure-to-Enroll (FTE) Rate

- A system's failure-to-enroll (FTE) rate represents the probability that a given user will be unable to enroll in a biometric system.
- FTEs occur when users **have insufficiently distinctive or replicable biometric data** or when the design of the biometric solution is such that providing consistent data is difficult.
- High failure-to-enroll rates can be **particularly problematic for a biometric system**, as users unable to enroll must verify through another biometric technology or authentication method.

Failure-to-Enroll by Technology

TECHNOLOGY	CAUSES OF FTE
Finger-scan	Low-quality fingerprints (scarred, faint, no breaks in ridges) Inconsistent submission (angle and pressure of placement) Difficult submission process (sliding, swiping finger)
Facial-scan	Glasses, hats, headwear Poor lighting Background composition
Voice-scan	Background noise Quality of acquisition device Inconsistent vocal samples (volume, duration, inflection)
Iris-scan	Difficult submission process (lining up eye with guides, gauging correct distance) Glasses (cause reflection)
Signature-scan	Inconsistent signatures
Hand-scan	Inability to position hand according to guide pegs
Retina-scan	Difficult submission process (must remain still for several seconds for each acquisition, multiple acquisitions required)
Keystroke-scan	Inconsistent typing patterns

Relation of Failure-to-Enroll to False Nonmatch Rate

- False match rates and false nonmatch rates are inversely related: Decrease one rate and the other rate increases, and vice versa.
- Adjusting system settings to reduce FTE rates can also affect false nonmatch rates.
- By reducing the number of distinctive or replicable features required for enrollment
- FTE rates can be lowered—marginal images and data can then be enrolled.

Derived Metrics

- **Equal Error Rate (EER)**: The equal error rate (EER), also referred to as the crossover rate, is the rate at which the FNMR is equal to the FMR.
- it represents the accuracy level at which likelihood of a false match is the same as the likelihood of a false non-match.
- Equal error rate is commonly used as a representation of overall system accuracy, as it is a general indicator of a system's resistance to break-ins and ability to match templates from authorized users

Derived Metrics

- EER is generated from single false match and single false nonmatch rates, neither of which is reflective of system operation in the real world.
- EER does not incorporate failure-to-enroll rates.
- A system might have an extremely low EER but an FTE rate of 15 percent.

Ability-to-Verify (ATV) Rate

- A more valuable derived metric is the **ability-to-verify (ATV) rate**.
- ATV is a combination of the failure-to-enroll and false nonmatch rates, and indicates the overall percentage of users who will be capable of authenticating on a daily basis.

$$ATV = (1 - FTE)(1 - FNMR)$$

➤ Assessing the privacy risks of biometrics

Designing privacy sympathetic biometric systems

Privacy Risks of Biometrics

- Unlike passwords, biometric traits are **permanent and difficult to change**. If compromised, an individual cannot simply replace their fingerprint or iris.
- **1. Irrevocability Risk**
 - Passwords can be changed if leaked.
 - Biometric traits cannot be easily changed.
 - Stolen biometric data may lead to lifelong identity risks.
- **Example:**

A fingerprint database is breached. Attackers may use the stolen fingerprints for spoofing systems elsewhere.

- **2. Unauthorized Tracking and Surveillance**

- Biometric systems can identify individuals without their knowledge.
- **Examples:**
 - Face recognition cameras in public spaces.
 - Tracking movement across shopping malls, airports, or cities.
- **Privacy Concern:**
 - Loss of anonymity.

3. Function Creep

- Data collected for one purpose may later be used for another purpose.
- **Example:**
A university collects facial data for attendance.
Later, the same data is used to monitor student behavior.

4. Data Breaches

- Biometric databases are attractive targets for attackers.
- **Consequences:**
 - Identity theft
 - Account compromise
 - Fraudulent authentication

5. Profiling and Discrimination

- Biometric data may reveal sensitive information:
- Age
- Gender
- Ethnicity
- Health conditions
- This information could be used for unfair decision-making.

6. Spoofing and Presentation Attacks

Attackers may use:
Fake fingerprints
Deepfake faces
Recorded voices
to bypass authentication systems.

Privacy Impact Assessment (PIA) for Biometrics

Question

What biometric data is collected?

Why is it collected?

Who can access it?

Where is it stored?

How long is it retained?

What happens after deletion?

What are the risks?

Assessment

Fingerprint, face, iris

Authentication

Admin, vendor

Local server/cloud

Defined retention period

Secure erasure

Leakage, surveillance

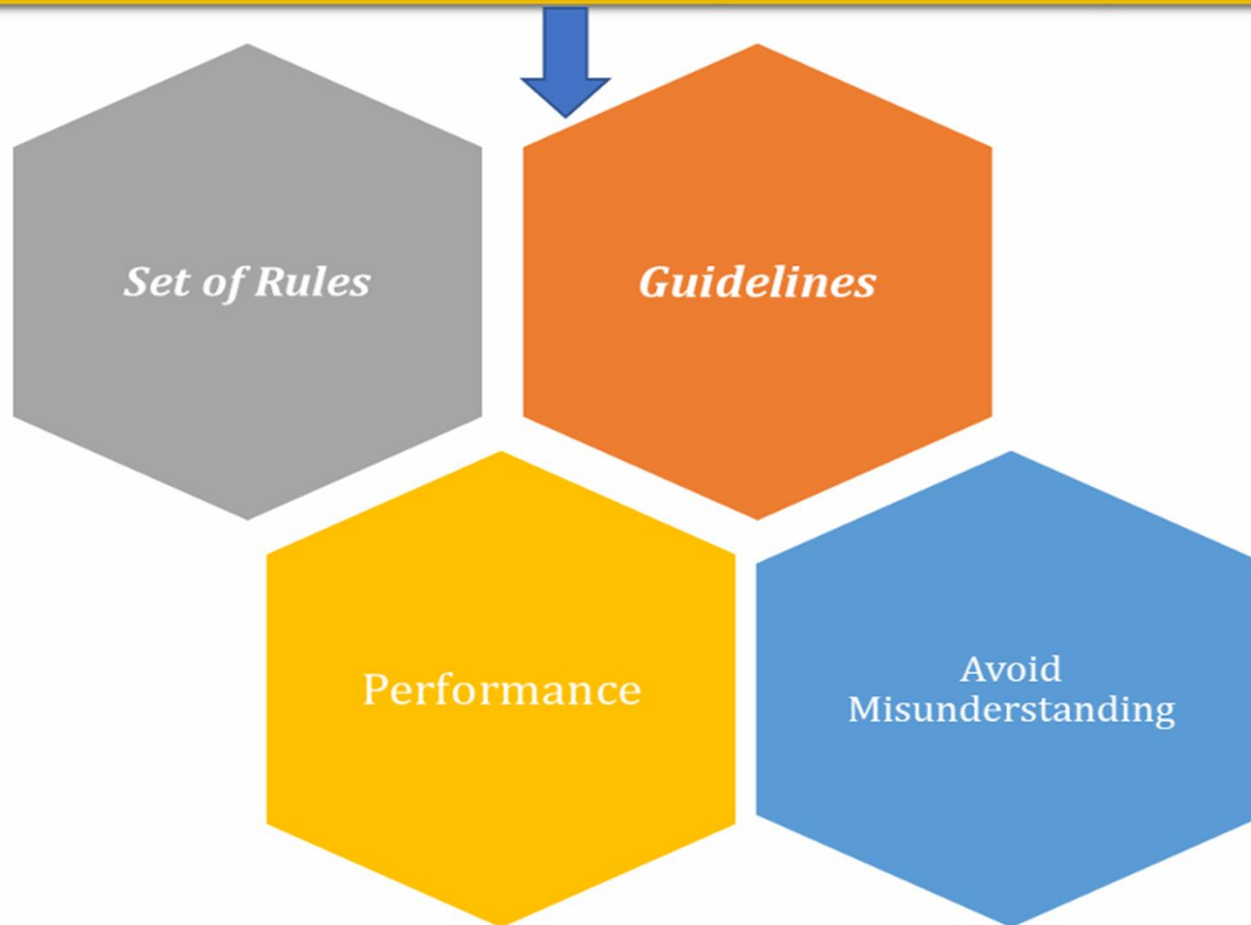
Designing Privacy-Sympathetic Biometric Systems

- A privacy-sympathetic biometric system protects users while still providing authentication services.
- **Principle 1: Data Minimization-** Collect only necessary biometric information.
- **Principle 2: *Template Protection***- *Store transformed biometric templates* instead of raw images.
 - **Techniques:** Hashing , Cancelable Biometrics ,Bio-Cryptography
 - **Benefit:**Even if stolen, original biometric traits cannot be reconstructed easily.
- **Principle 3: Encryption, User Consent and Transparency**
- Decentralized Storage

Different biometric standards,

- The Biometric Consortium website defines standards as “a general set of rules to which all complying procedures, products or research must adhere.”
- Standards play a role in everyday life by establishing the size, configuration, or protocol of a product, process, or system.

Standards: *"a document, established by consensus that provides:*



- International Organization for Standardization (ISO)
- International Electrotechnical Commission (IEC)
- **Responsibilities:**
 - Biometric data formats
 - Performance evaluation
 - Security requirements
 - Privacy guidelines
 - Testing methodologies
- **Examples**
 - ISO/IEC 19794
 - ISO/IEC 24745
 - ISO/IEC 30107

2. NIST

- NIST- National Institute of Standards and Technology
- NIST develops:
 - Biometric testing frameworks
 - Accuracy benchmarks
 - Face Recognition Vendor Tests (FRVT)
 - Fingerprint Vendor Technology Evaluation (FpVTE)
- **Contribution**
- Evaluates biometric algorithms and provides performance benchmarks.

- Standards for the collection, storage, and sharing of biometric data are of utmost importance to government and private systems.
- International Organization for Standardization (ISO)/International Electrotechnical Commission (IEC) Guide 2:2004 defines a standard as “a document, established by consensus that provides rules, guidelines or characteristics for activities or their results.”
- The Biometric Consortium website defines standards as “a general set of rules to which all complying procedures, products or research must adhere.”
- Standards play a role in everyday life by establishing the size, configuration, or protocol of a product, process, or system.
- Standards specify performance of products or personnel and also define terms so that there is no misunderstanding among those using the standards

Biometric standards specify:

formats for the
interchange of
biometric data

common file formats
that provide platform
independence

application program
interfaces and
application profiles

performance metric
definitions and
calculations

approaches to test
performance

Technical Interfaces:

- specify interfaces and interactions between biometric components and subsystems, including the possible use of security mechanisms to protect stored data and data transferred between systems.
- Examples include ANSI INCITS 358-2002 BioAPI Specification v1.1 and ANSI INCITS 398-2005 [NISTIR 6529-A] Common Biometric Exchange File Format (CBEFF).

Data Interchange Formats:

- Examples
- ANSI INCITS 377-2004 Finger Pattern Based Interchange Format
- ANSI INCITS 378-2004 Finger Minutiae Format for Data Interchange,
- ANSI INCITS 379-2004 Iris Image Interchange Format.

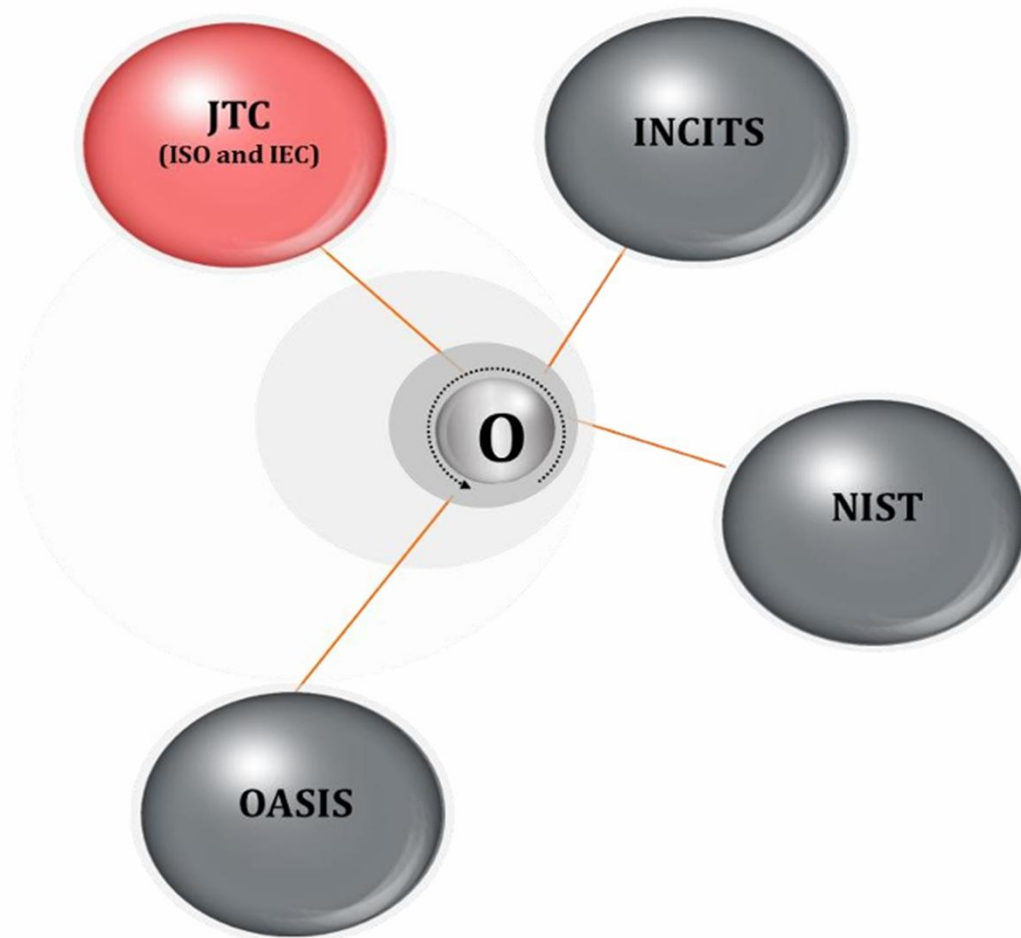
Application Profile Standards:

- Examples include ANSI INCITS 383-2003 Biometrics-Based Verification and Identification of Transportation Workers, and ANSI INCITS 394-2004 Data Interchange and Data Integrity of Biometric-Based Personal Identification for Border. Management.

Performance Testing and Reporting :

- Examples include ANSI INCITS 409.1-2005 Biometric Performance Testing and Reporting Part 1 - Principles Framework; ANSI INCITS 409.2-2005 Biometric Performance Testing a Reporting Part 2 - Technology Testing Methodology; and ANSI INCITS 409.3-2005 Biometric Performance Testing and Reporting Part 3 - Scenario Testing Methodologies.

Major Organizations involved in drafting Standards



- **INCITS:** InterNational Committee for Information Technology Standards (INCITS)
- **NIST:** National Institute of Standards and Technology
- **JTC:** Joint Technical Committee 1 (JTC 1)/Subcommittee 37 (SC 37). ISO and IEC Joint Technical Committee
- **OASIS:** Organization for the Advancement of Structured Information Standards.

Working groups of JTC

Working Group	Responsibilities
WG1 Harmonized Biometric Vocabulary	develops standardized definitions for biometric vocabulary terms
WG2 Biometric Technical Interfaces	develops international standards for BioAPI and CBEFF, as well as a number of other related projects.
WG3 Biometric Data Interchange Formats	develops international versions of the biometric data interchange format standards.
WG4 Biometric Functional Architecture and Related Profiles	develops international biometric profile standards to support biometric interoperability for applications.
WG5 Biometric Testing and Reporting	develops international standards for biometric performance testing and reporting.
WG6 Cross jurisdictional and Societal Aspects	currently developing an international technical report on privacy concerns and other social concerns related to biometric standards.